

Exhibit 2

EXHIBIT 3

Oura Ring 4 Infringement Claim Chart - U.S. Patent No. 10,231,675

	U.S. Patent No. 10,231,675
Independent Claim 13	Oura Ring 4 and Corresponding Applications (collectively, “Oura Accused Product”)
13[pre] 13. A method comprising:	To the extent that the preamble is limiting, the Oura Accused Product comprises a method.
13[a] determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device;	The Oura Accused Product includes determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device. The Oura Accused Product includes a sensor, such as an accelerometer, for determining whether is the electronic ring is worn on a part of a user’s body, such as a finger. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep.

This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.

**EXTENDED BATTERY LIFE**

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

**DATA PROCESSING**

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))

Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

(SAMS382-0027802, <https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4> (retrieved March 10, 2026))

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#). (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
13[b] acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body;	The Oura Accused Product includes acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body. The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time, such as the period of time shortly before or immediately after it is determined that the electronic device is worn on a part of the user's body, where the first period of time is a period of time when the electronic ring is not being worn. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

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	U.S. Patent No. 10,231,675
13[c] activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;	The Oura Accused Product includes activating at least one second sensor based on at least a part of the first information on the motion of the electronic device. The Oura Accused Product activates at least one second sensor on the electronic ring device, such as a red and/or infrared (IR) LED and sensor, based on at least a part of the first information on the motion of the electronic device. The red and/or infrared (IR) LED could be activated after the first information on the motion indicates that the user being asleep or the motion signal satisfies a condition indicative of sleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

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How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

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Deep sleep:

- Low movement
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REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(SAMS382-0028038, <https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages> (retrieved March 10, 2026))

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(SAMS382-0027774, <https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
13[d] detecting a biometric information of the user using	The Oura Accused Product includes detecting a biometric information of the user using the activated at least one second sensor.

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the activated at least one second sensor;	The Oura Accused Product detects biometric information of the user, such as blood oxygen (SpO2) levels, using the activated at least one second sensor, such as the red and/or infrared (IR) LED and sensor. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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13[e] during detecting the biometric information,	The Oura Accused Product includes, during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor.

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acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and	While detecting the biometric information, such as blood oxygen (SpO2) levels, the Oura Accused Product acquires second information on a motion of the electronic device using at least the part of the at least one first sensor, such as the accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

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13[f] in response to identifying that the motion of the	The Oura Accused Product includes, in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

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electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.	The Oura Accused Product can identify that the motion of the electronic ring device included in the second information is more than a threshold motion, which could signal that the user is not asleep or the second information satisfies a condition indicative of not being asleep. In response to determining that the user is not asleep based on the user's motion, the Oura Accused Product can then inactivate the at least one second sensor, such as the red and/or infrared (IR) LED and sensor. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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(SAMS382-0027774, <https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
Dependent Claim 14	

	U.S. Patent No. 10,231,675
14. The method of claim 13, wherein the at least one first sensor comprises at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor.	The Oura Accused Product includes wherein the at least one first sensor comprises at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor. The Oura Accused Product includes at least one acceleration sensor, such as an accelerometer. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675	
	Sensors • Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))
Dependent Claim 15	
15[a] The method of claim 13, further comprising: acquiring additional information on the motion of the electronic device using at least a part of the at least one	The Oura Accused Product includes acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes. The Oura Accused Product, during a third period of time beginning immediately after the second period of time (e.g., the period of time ending after the user wakes up), which may be shorter than a first period of time when

	U.S. Patent No. 10,231,675
first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and	the Galaxy Watch product is not being worn, acquires additional information on the motion of the electronic device using the at least one first sensor, e.g., an accelerometer. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
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15[b] activating the at least one second sensor based on at least a part of the additional information.	The Oura Accused Product includes activating the at least one second sensor based on at least a part of the additional information. The Oura Accused Product activates the one second sensor, such as a red and/or infrared (IR) LED and sensor, based on at least a part of the additional information on the motion of the electronic ring device during the third time period, such as the user being asleep or the motion signal satisfying a condition indicative of sleep.		

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(SAMS382-0027592, <https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
Dependent Claim 16	
16. The method of claim 15, wherein at least one of the second period of time or the third period of time comprises a time point when the first period of time passes.	The Oura Accused Product includes wherein at least one of the second period of time or the third period of time comprises a time point when the first period of time passes. For example, the Oura Accused Product can, during a second period of time The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time, such as the period of time when it is determined that the electronic device is worn on a part of the user's body, shortly before and/or immediately after a first period of time, such as a period of time when the electronic ring is not being worn. The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time shortly before it is determined that the electronic device is worn on a part of the user's body, which comprises a time point when the first period of time passes <i>See Claims 13 and 15. See also:</i>

	U.S. Patent No. 10,231,675
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
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	U.S. Patent No. 10,231,675
Dependent Claim 17	
17[a] The method of claim 13, further comprising: accumulating a number of failures when the detection of the biometric information fails; and	The Oura Accused Product includes accumulating a number of failures when the detection of the biometric information fails. The Oura Accused Product can detect and accumulate failures to detect biometric information from one or more biological sensors such as red and/or infrared (IR) LEDs and sensors. For example, the Oura Accused Product includes Smart Sensing, that adjusts which sensors to activate based on factors such as finger structure, skin tone, activity, and ring orientation, which includes deactivating sensors based on failure to obtain biometric information. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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			(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))
17[b] deactivating the at least one second sensor when a cumulative number of failures satisfies a selected condition.	The Oura Accused Product includes deactivating the at least one second sensor when a cumulative number of failures satisfies a selected condition. The Oura Accused Product, including the Smart Sensing feature, deactivates sensors such as the red and/or infrared (IR) LEDs and sensors when a cumulative number of failures satisfies a selected condition.		

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<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

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Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(SAMS382-0027592, <https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
Dependent Claim 18	
18[a] The method of claim 13, wherein the detection of the biometric information of the user using the activated at least one second sensor comprises: acquiring data about a physical characteristic or state of the user; and	The Oura Accused Product includes wherein the detection of the biometric information of the user using the activated at least one second sensor comprises acquiring data about a physical characteristic or state of the user. The Oura Accused Product includes Smart Sensing, that detects data about a physical characteristic or state of the user, such as blood oxygen (SpO2), heart rate, and respiration rate. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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18[b] changing at least one condition for activating the at least one second sensor based on at least a part of the data.	The Oura Accused Product includes changing at least one condition for activating the at least one second sensor based on at least a part of the data. The Oura Accused Product includes Smart Sensing, which adjusts which sensors to activate based on factors such as finger structure, skin tone, activity, and ring orientation, which includes changing at least one condition for activating the at least one second sensor based on at least a part of the data about a physical characteristic or state of the user.		

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	U.S. Patent No. 10,231,675
Dependent Claim 19	
19[a] The method of claim 13, wherein the detection of the biometric information of the user using the activated at least one second sensor comprises: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and	The Oura Accused Product includes wherein the detection of the biometric information of the user using the activated at least one second sensor comprises acquiring data from at least one of the at least one first sensor and the at least one second sensor. The Oura Accused Product acquires motion data from at least one of the at least one first sensor, a motion sensor such as an accelerometer, and biometric data, such as SpO2 data, from the at least one second sensor, such as red and/or infrared (IR) LEDs and sensors. <i>See Claim 13. See also:</i> Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675			
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	U.S. Patent No. 10,231,675
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(SAMS382-0027592, <https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
Independent Claim 20	
20[pre] 20. An electronic device comprising:	To the extent that the preamble is limiting, the Oura Accused Product includes an electronic device, specifically a wearable electronic ring. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))

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A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	Hardware Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold. Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide. All Oura Ring 4 models are available in sizes 4-15. The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit. • Oura Ring 4 Sizing Kit: white ring sizes, sizes 4–15 • Oura Ring Gen3 Sizing Kit: black ring sizes, sizes 6–13 For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger. (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#). (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
20[a] at least one first sensor;	The Oura Accused Product includes at least one first sensor. The Oura Accused Product includes a first sensor that is a motion sensor, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4 (retrieved March 10, 2026))

U.S. Patent No. 10,231,675	
	Sensors • Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))
20[b] at least one second sensor;	The Oura Accused Product includes at least one second sensor. The Oura Accused Product includes a second sensor, such as a red and/or infrared (IR) LED and sensor.

	U.S. Patent No. 10,231,675
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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			(SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

(SAMS382-0027769, <https://support.ouraring.com/hc/en-us/articles/4410651298963-Live-Heart-Rate> (retrieved March 10, 2026))

Oura Ring 4: More Accurate Than Ever Before

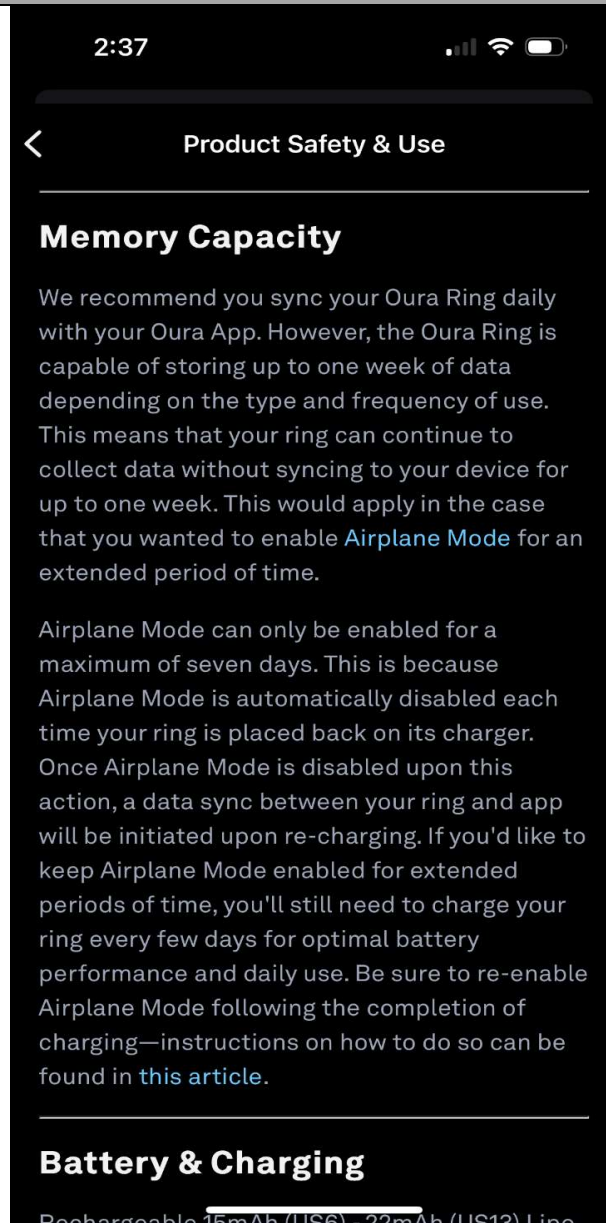


Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

	U.S. Patent No. 10,231,675
20[c] a memory; and	The Oura Accused Product includes a memory. The Oura Accused Product includes memory, such as RAM and storage.



(Oura App → Help → Product Safety & Use → Memory Capacity)

	U.S. Patent No. 10,231,675
	Memory Capacity We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable Airplane Mode for an extended period of time. Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in this article. (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))
20[d] at least one processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory,	The Oura Accused Product includes at least one processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. The Oura Accused Product comprises an Infineon PSoC 6 Controller that is connected to, among other things, the accelerometer, the red and/or infrared (IR) LED and sensor, RAM, and storage.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep.

This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.

**EXTENDED BATTERY LIFE**

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

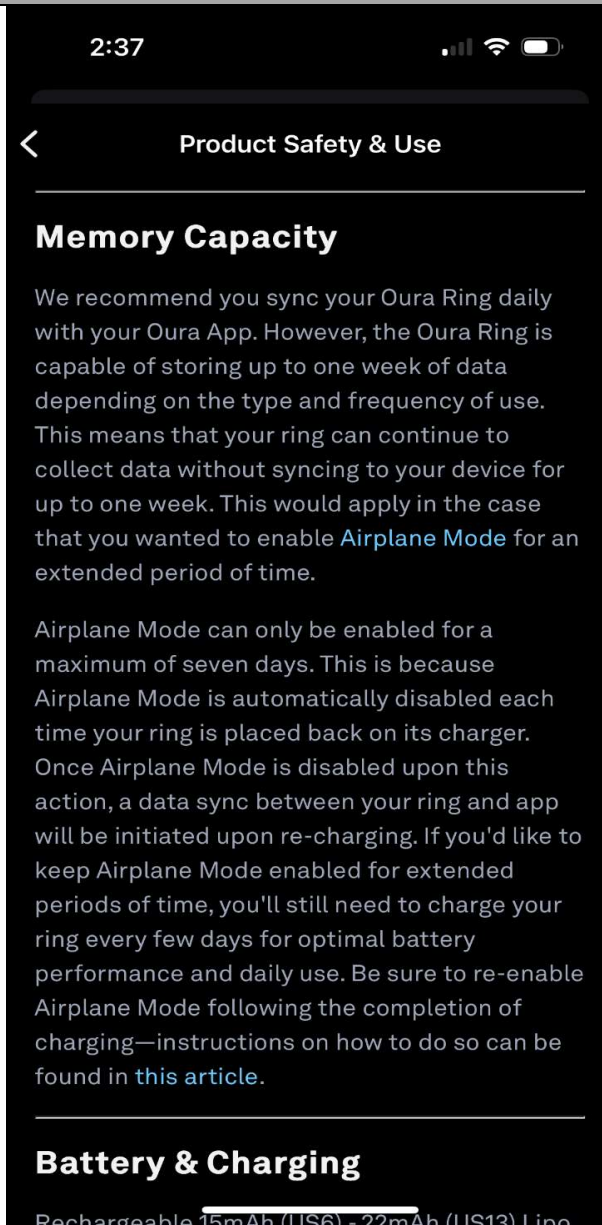
SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

**DATA PROCESSING**

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))



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20[e] wherein the memory stores one or more computer programs including instructions which, when executed by the at least one processor, cause the at least one processor to:	The Oura Accused Product includes a memory that stores one or more computer programs including instructions which, when executed by the at least one processor, cause the at least one processor to perform certain tasks, which are described below in limitations [e][i] – [e][vi]).

	U.S. Patent No. 10,231,675
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 16 and higher, or Android devices running Android 9 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

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SENSOR FUSION

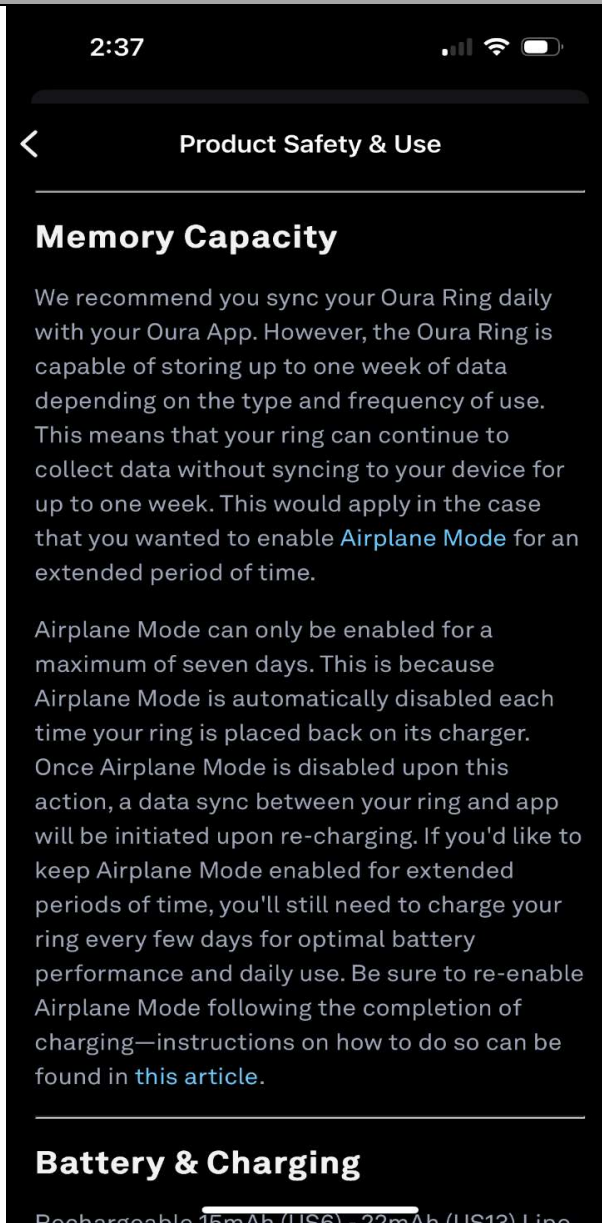
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(SAMS382-0028055, <https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf> (retrieved March 10, 2026))



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20[e][i] determine whether the electronic device is worn on a part of a user's body,	The Oura Accused Product determines whether the electronic device is worn on a part of a user's body. The Oura Accused Product includes memory that causes the processor to determine whether the electronic device is worn on a part of a user's body, such as a finger, using sensor(s) for wear detection.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

(SAMS382-0027612, <https://ouraring.com/blog/oura-ring-4/> (retrieved March 10, 2026))

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	Hardware Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold. Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide. All Oura Ring 4 models are available in sizes 4-15. The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit. • Oura Ring 4 Sizing Kit: white ring sizes, sizes 4–15 • Oura Ring Gen3 Sizing Kit: black ring sizes, sizes 6–13 For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger. (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

General Use

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We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

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For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#). (SAMS382-0027816, https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
20[e][ii] when the electronic device is worn on the part of the user's body, determine a motion of the electronic device based on first information obtained by the at least one first sensor,	The Oura Accused Product, when the electronic device is worn on the part of the user's body, determine a motion of the electronic device based on first information obtained by the at least one first sensor. The Oura Accused Product includes memory that causes the processor to determine, when the electronic ring device is worn on the user's finger, a motion of the electronic ring device obtained by the at least one first sensor, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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	U.S. Patent No. 10,231,675
	https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y
20[e][iii] determine whether to identify biometric information based on the motion of the electronic device,	The Oura Accused Product determines whether to identify biometric information based on the motion of the electronic device. The Oura Accused Product includes memory that causes the processor to determine whether to identify biometric information, such as blood oxygen (SpO₂) levels, based on the motion of the electronic device, such as the user being asleep or the motion signal satisfying a condition indicative of sleep. In particular, when the motion signal indicates the user may be sleeping or is asleep, Red and infrared LEDs measure blood oxygen, SpO₂. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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(<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

	U.S. Patent No. 10,231,675
	(SAMS382-0028038, https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages (retrieved March 10, 2026)) How Nap Detection Works For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one sleep stage (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the 24-hour heart rate graph as restorative time (green). Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically. (SAMS382-0027774, https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
20[e][iv] in response to determining to identify	The Oura Accused Product, in response to determining to identify biometric information, identifies the biometric information of the user through the at least one second sensor.

	U.S. Patent No. 10,231,675
biometric information, identify the biometric information of the user through the at least one second sensor,	The Oura Accused Product includes memory that causes the processor to identify biometric information, such as blood oxygen (SpO₂) levels, through the use of red and/or infrared (IR) LED and sensor, in response to determining that the user is asleep or the motion signal satisfies a condition indicative of sleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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	(SAMS382-0028038, https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages (retrieved March 10, 2026)) How Nap Detection Works For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one sleep stage (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the 24-hour heart rate graph as restorative time (green). Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically. (SAMS382-0027774, https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection (retrieved March 10, 2026))

	U.S. Patent No. 10,231,675
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (SAMS382-0027592, https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2 (retrieved March 10, 2026))
20[e][v] during identifying the biometric information of the	The Oura Accused Product, during identifying the biometric information of the user, obtain second information on a motion of the electronic device using the at least one first sensor.

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user, obtain second information on a motion of the electronic device using the at least one first sensor, and	The Oura Accused Product includes memory that causes the processor to obtain second information corresponding to the motion of the electronic device using the at least one first sensor, such as an accelerometer. Such motion information is obtained while biometric information, such as blood oxygen (SpO₂) levels is obtained, such as when the user is asleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (SAMS382-0027802, https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4 (retrieved March 10, 2026))

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Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

(<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

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20[e][vi] in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivate the at least one second sensor.	The Oura Accused Product, in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivates the at least one second sensor. The Oura Accused Product includes memory that causes the processor to inactivate the at least one second sensor, such as the red and/or infrared (IR) LED and sensor, in response to identifying that the motion of the electronic ring device included is more than a threshold level of motion. When the user's motion exceeds this threshold level, it could, for example, signal that the user is not asleep or the second information about motion satisfies a condition indicative of not being asleep.

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